

Towards Optimizing Urban Regeneration in Post-Industrial Cities via Computational Analysis of Digitized Urban Structures

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This paper introduces a computational framework to evaluate urban structure for optimizing regeneration efforts in post-industrial cities through the spatial analysis of digitized urban structures. By identifying five key structural factors—urban form, street configuration, residential buildings, street networks, and urban blocks—and 24 associated indicators, the study proposes a method for evaluating the spatial potential of regeneration sites. The framework employs GIS, space syntax, and network analysis to assess how urban structure influences the diffusion of socio-economic and environmental benefits to surrounding areas. Importantly, the research contributes to ambiance-oriented urban studies by linking spatial configurations with the capacity to shape sensory experiences, social interactions, and the perceived quality of urban life. This integration of computational precision with ambiance-sensitive planning supports a more holistic and evidence-based approach to urban transformation. The proposed methodology offers a replicable tool for designers, planners, and policymakers aiming to prioritize regeneration projects that foster inclusive, vibrant, and experientially rich urban environments.

Keywords: Urban regeneration, Ambiance-oriented urban structure analysis, Computational framework, Space syntax

INTRODUCTION

Urban regeneration focuses on addressing urban decay, involving diverse measures like land reutilization, brownfield redevelopment, and cultural and social improvements tailored to specific historical and national contexts (Figueiredo et al., 2022; Xie et al., 2021). In fact, it is defined as a process to improve the urban fabric and meet economic, environmental, and social objectives, incorporating a comprehensive vision to solve urban problems and generate long-term transformations (Shi et al., 2023).

Urban regeneration has transitioned from a primary focus on physical redevelopment to a more integrated approach that incorporates

economic renewal, social inclusion, and environmental sustainability. Initially, regeneration efforts aimed at improving deteriorating urban areas, but their scope has expanded to include enhancing livability and , fostering active community participation (Said & Dindar, 2024). Modern strategies prioritize participatory governance, emphasizing the role of local communities in shaping sustainable urban transformations (Heath et al., 2023).

However, literature reviews indicate that existing studies have seldom employed quantitative and computational analyses to enhance the efficiency of urban regeneration across various domains. One critical domain

involves the spatial assessment of urban regeneration project locations within cities based on their social, economic, and environmental impacts on their surrounding context.

To more efficiently impact their surrounding context, urban regeneration projects should have the ability to motivate social and economic capital to utilize their patterns and become integrated into them (Ferilli et al., 2017). The built environment appears to play a crucial role in facilitating these movements (Li et al., 2022), and urban structure, defined as the spatial organization of land uses and transport networks that shape how a city functions (Pacione, 2009), has been shown to stimulate potential social and economic activities generated by that urban projects such as regeneration projects and spread them throughout the surrounding urban context (McFarlane et al., 2023; Mohamed et al., 2022; Webber, 1964). Given that the primary goal of efficient urban regeneration projects is to accelerate the revitalization of their surrounding urban context across environmental, social, and economic dimensions (Ferilli et al., 2017), research suggests that well-structured, accessible, and integrated urban street networks facilitate this process.

However, previous studies have rarely addressed the critical role that urban structure can play in influencing the overall effectiveness of urban regeneration projects. In particular, there has been limited discussion on how the spatial configuration of land uses and transport networks might contribute to the diffusion of a regeneration project's social and economic benefits beyond its immediate boundaries (Chakraborty & Fukuda, 2025). This omission overlooks the potential for the urban structure to act as a catalyst, facilitating the transmission of positive outcomes—such as increased accessibility, economic revitalization, and social cohesion—to other areas within the broader urban context. Furthermore, the methodological frameworks employed in many of these studies

often lack the integration of computational tools that are essential for analyzing spatial relationships and network effects. Without such tools, it remains challenging to quantify the extent to which regenerating a specific urban area might influence its surrounding neighborhoods and contribute to broader urban transformation.

In light of the preceding discussion, the primary research objective of this study is twofold. First, we aim to identify potential computational parameters within urban structures that can enhance and stimulate social, economic, and environmental activities in the surrounding areas of urban regeneration projects. Specifically, this study seeks to determine which computational urban structure parameters can serve as criteria for identifying the optimal locations for urban regeneration in post-industrial cities where multiple options exist. Second, the study aims to develop a computational framework of indicators and define how these indicators can be measured and applied in practice.

To identify the relevant parameters and their measurement methods, this study conducts a comprehensive review of key concepts, including urban regeneration projects, urban structure components, and their analytical framework. Based on the synthesized findings, a computational model integrating optimized urban structure criteria is formulated to enhance the effectiveness of urban regeneration projects.

These computational parameters are expected to be particularly significant for post-industrial European cities, where multiple urban regeneration projects compete for limited financial resources. Given these budgetary constraints, optimizing project selection through a set of computationally derived criteria constitutes a key contribution of this study.

URBAN STRUCTURE PARAMETERS AND ITS MEASUREMENT TOOLS

A review of the existing literature reveals that various sets of factors and indicators influence the

structure of urban environments. These factors are often context-dependent and interact in complex ways. Therefore, in order to develop a more comprehensive understanding of urban structure, this section outlines the key influencing factors along with their corresponding sub-indicators.

The most emphasized urban structural factor, frequently mentioned in the literature, is urban form, which refers to the physical configuration of urban areas, including the organization of streets, buildings, public spaces, and supporting infrastructure (Monteiro et al., 2024; X. Zhang et al., 2024). Its key indicators include Dimension, Shape, Spatial distribution, Intensity, Connectivity, and Diversity. These can be measured using metrics (e.g., shape index, entropy, FAR) and toolssuch as GIS software (e.g., ArcGIS, QGIS), spatial statistics (e.g., NNA), and space syntax software (e.g., DepthmapX) (Fleischmann et al., 2021; Hillier et al., 1993). . Table 1 presents these six indicators along with their descriptions.

Table 1
Urban form indicators and definitions (Fleischmann et al., 2021; Hillier et al., 1993)

ID	Urban form's Indicators	Description
1	Dimension	The basic geometrical dimensions of individual objects
2	Shape	The mathematical features of geo-metrical dimensions of individual objects
3	Spatial distribution	The spatial distribution of objects in space and their reciprocal positioning
4	Intensity	The Intensity of space occupation, referring to the density of elements within a set context
5	Connectivity	The spatial interconnection of the segments of the networks (usually street networks)
6	Diversity	The variety and richness of the elements and their characteristics in the study area

Table 2
Street configuration indicators and definitions (Fleischmann et al., 2021; Tannous et al., 2021; Weber et al., 2014).

The second most well-known factor for measuring urban structure is street configuration, which refers to how streets and intersections are organized spatially, shaping the connectivity and overall layout of an urban street network (Li et al., 2022). According to the literature, it has three main indicators: global integration, local integration, and global choice, which are measured using spatial network analysis based on graph theory, typically calculated with space syntax tools such as DepthmapX or UCL Depthmap (Fleischmann et al., 2021; Tannous et al., 2021; Weber et al., 2014).

ID	Street configuration's indicators	Description
1	Global integration	spaces with higher levels of global integration tend to carry higher levels of movement and, hence, a greater potential to access different varieties of land use
2	Local integration	Local integration measures relativized mean depth to three (3) changes of direction away from an origin space. It is a more immediate measure of the local catchment area of a single space.
3	Global choice	Global choice is a measurement of 'through-movement' based on giving every space in the spatial network (however represented) a value of 1, then proportionally sharing that value amongst all its immediate connections.

ID	Residential building's indicators	Description
1	Dimension	Mean area of residential buildings
2	Shape	Mean fractal dimension index for residential buildings
3	Intensity	Edge density for residential buildings
4	Spatial distribution	Average Euclidean nearest-neighbor distance for residential buildings
5	Connectivity	Percentage of shared edges in total edge length

The third factor that is proven to affect urban structure is the types of residential buildings in a specific urban context, such as single-family homes, apartments, condominiums, and townhouses (Weerasekara et al., 2023). Within the framework of urban structure, a residential building is defined as a housing unit whose morphological attributes—including Dimension, Shape, Intensity, Spatial Distribution, and Connectivity (Table 3)—constitute critical determinants of the spatial configuration and functional organization of the urban fabric, and can be quantitatively measured using spatial metrics and geospatial analysis tools such as GIS (e.g., ArcGIS, QGIS) and landscape metrics software (e.g., FRAGSTATS) (Fleischmann et al., 2021; Łaskiewicz et al., 2022; Ma et al., 2020; Weber et al., 2014; Wu & Murray, 2008).

In the context of urban structure, street network indicators—considered the fourth fundamental component influencing urban structure—are quantitative descriptors of the geometric and topological features of the street system (Boeing, 2020). It encompasses metrics such as Dimension, Shape, Spatial distribution, Intensity, and Connectivity (Table 4) and can be quantitatively measured using graph-based

spatial network analysis, employing tools like GIS software (e.g., ArcGIS, QGIS) and urban network analysis libraries (e.g., OSMnx, NetworkX). These methods allow for the extraction, modeling, and computation of both geometric and topological properties of street systems (Boeing, 2018; Łaskiewicz et al., 2022). Also, these indicators play a critical role in urban structure analysis, as they govern patterns of movement, accessibility, and the spatial logic underlying urban functionality.

ID	Street network's Indicators	Description
1	Dimension	Average street segment length
2	Shape	Average circuitry, the ratio of network distances to straight-line distances
3	Spatial distribution	Average closeness centrality (Boeing 2018)
4	Intensity	Street density
5	Connectivity	Weighted average clustering coefficient

Last but not least, the urban block factor is a core spatial unit defined by the perimeter of intersecting streets and occupied by built forms (Fleischmann et al., 2021), whose morphological characteristics—such as Dimension, Shape, Spatial distribution, Intensity, and Connectivity—play a critical role in shaping the urban structure. (Fleischmann et al., 2021; Łaskiewicz et al., 2022; Oliveira et al., 2020; Z. Zhang et al., 2019). These urban block indicators are quantified through the use of spatial data analysis, statistical modeling, and geometric calculations, employing tools such as GIS, FRAGSTATS, R, (Lowry & Lowry, 2014). (Table 5).

Table 3
Residential building indicators and definitions (Fleischmann et al., 2021; Łaskiewicz et al., 2022; Ma et al., 2020; Weber et al., 2014; Wu & Murray, 2008)

Table 4
Street networks indicators and definitions (Boeing, 2018; Łaskiewicz et al., 2022)

Table 5
Urban block
indicators and
definitions
(Fleischmann et
al., 2021;
Łaskiewicz et al.,
2022; Oliveira et
al., 2020; Z. Zhang
et al., 2019)

ID	Urban block's Indicators	Description
1	Dimension	Urban block area
2	Shape	Fractal dimension index for urban block
3	Spatial distribution	Squared Euclidean distance between urban block and its neighbors in terms of building density
4	Intensity	Number of residential buildings per urban block area
5	Connectivity	Number of adjacent urban blocks

Based on the above review, five main factors contributing to the construction of urban structures have been identified: urban form, street configuration, residential building types, street networks, and urban blocks. Additionally, their measurement tools and methods have been briefly discussed.

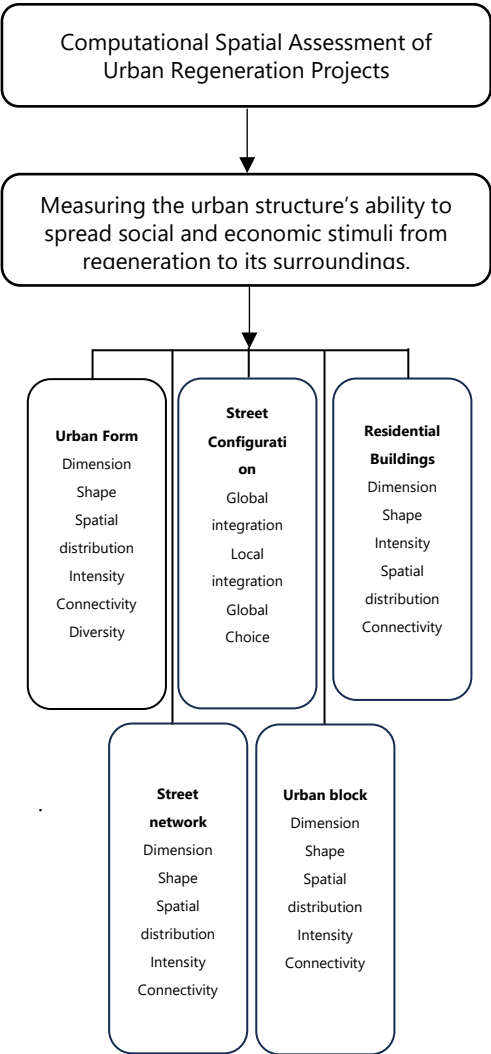
ENHANCING URBAN REGENERATION IMPACTS THROUGH COMPUTATIONAL ANALYSIS OF URBAN STRUCTURE

Following the transformation of several European cities from industrial-based societies and economies to post-industrial ones, a substantial number of former industrial sites have become obsolete and remain without functional use throughout these urban areas. Despite the increasing need to reintegrate these underutilized spaces into the urban fabric, limitations in financial resources and managerial capacities often restrict the scope of regeneration initiatives. Consequently, it is not feasible to regenerate all abandoned industrial spaces simultaneously. Therefore, it becomes essential to establish a set of objective and systematic criteria to assess and prioritize which sites hold the highest potential

and should be considered as suitable candidates for regeneration interventions.

One of the key criteria for identifying the most promising spaces for urban regeneration is the capacity of the urban structure—where the potential regeneration project is located—to facilitate the diffusion of social and economic development throughout the surrounding urban context. In fact, it is essential to address the question: what types of urban structures are most effective in transmitting the social and economic benefits generated by regeneration projects to other parts of the city? To address this question, this study reviews the key factors that explain urban structure and identifies the indicators used to measure each factor. Additionally, the computational tools applicable to each indicator are introduced.

Figure 1 presents the five key factors and their 24 associated indicators that together constitute the fundamental components of urban structure. These indicators should be systematically measured and evaluated, particularly when urban regeneration projects in post-industrial cities are being planned, in order to ensure that decisions are made not only based on broader economic, social, or environmental criteria but also with a clear understanding of the existing urban fabric. In fact, considering these computational spatial factors enhances the likelihood of extending the social and economic stimuli of urban regeneration projects to their surrounding context.



Across most of the five factors, indicators such as dimension, shape, connectivity, spatial distribution, and intensity are the most frequently used to measure various aspects of urban structure, including urban form, residential

building types, street networks, and urban blocks. In addition, the most prominent methods for measuring these indicators involve spatial statistics, space syntax, and network analysis, which are typically conducted using specialized software tools such as ArcGIS, QGIS, DepthmapX, and FRAGSTATS, allowing for detailed spatial, morphological, and topological analyses of urban environments.

DISCUSSION

In recent years, the rise of data-driven approaches has significantly reshaped the way cities are understood, planned, and governed. By leveraging computational tools, geospatial data, and quantitative analysis, these methods offer new insights into the complex dynamics of urban systems, enabling more precise evaluations of spatial efficiency and urban form. This shift not only enhances the technical capacity to assess and optimize urban environments but also redefines the roles of key actors in the development process. Urban planners, policymakers, and stakeholders are increasingly required to interpret and act upon data-informed models, which can democratize decision-making but also raise questions about technocratic control and algorithmic governance. Consequently, data-driven approaches are not merely technical innovations—they represent a paradigm shift in how cities are conceptualized and how responsibility for their transformation is distributed among various institutional and civic agents. However, this shift also brings to light a growing tension between algorithmic optimization—which prioritizes quantifiable efficiency—and the more emergent, participatory processes of regeneration that rely on local knowledge, contested values, and community-driven priorities.

To develop a data-driven and quantitative optimization framework aimed at improving the efficiency of urban regeneration projects, five key

Figure 1
Computational
process for
identifying the
optimal candidate
space for
regeneration
through the
assessment of
urban structures
(Authors, 2025)

factors were identified and extracted through an extensive review of the relevant literature. These factors were chosen based on their prominence and frequency across multiple studies, as well as their proven impact on the spatial and functional dynamics of urban environments. The selection process involved a careful synthesis of existing theoretical and empirical works, with particular attention paid to indicators that consistently appeared as determinants of successful regeneration efforts.

The categorization of influential factors in the literature revealed that five physical-spatial components—urban form, street configuration, residential building layout, street network structure, and urban block arrangement—collectively play a critical role in shaping the outcomes of urban regeneration initiatives. These elements were found to significantly influence accessibility, connectivity, density, and overall spatial logic, all of which contribute to the operational efficiency and long-term success of regeneration projects. In contrast, other environmental and physical variables, although occasionally mentioned, were found to have a relatively limited effect and were therefore considered secondary in the proposed optimization framework.

While this study emphasizes the importance of spatial and structural factors, it is important to acknowledge that non-spatial dimensions also play a decisive role in the success or failure of urban regeneration projects. Factors such as political will, governance structures, community engagement, cultural identity, and economic feasibility often shape the practical implementation and long-term sustainability of regeneration efforts. Although these aspects were not the primary focus of this research due to its methodological emphasis on measurable spatial components, they should be seen as complementary and critical to achieving holistic and context-sensitive regeneration outcomes. Future studies would benefit from integrating

these socio-political and economic variables to further enrich the optimization framework.

CONCLUSION

This study contributes to the scientific community by offering a novel, systematic framework for integrating computational spatial analysis into urban regeneration planning. While previous research has largely focused on qualitative or case-based approaches, this study bridges a critical methodological gap by operationalizing 24 structural indicators across five core urban components—urban form, street configuration, street networks, residential buildings, and urban blocks—using quantitative tools such as GIS, space syntax, and network analysis. The proposed framework enables a data-driven assessment of a site's potential to spread regeneration benefits, thereby advancing the field of urban planning with an evidence-based methodology for prioritizing regeneration projects in resource-constrained, post-industrial contexts. This interdisciplinary contribution supports more effective spatial decision-making and sets a foundation for future comparative studies across different urban systems.

Furthermore, the research demonstrates that indicators such as dimension, shape, intensity, spatial distribution, and connectivity are consistently relevant across all structural factors. These indicators can be quantitatively assessed using spatial statistics, network analysis, and space syntax methods, supported by tools like ArcGIS, QGIS, DepthmapX, and FRAGSTATS.

Ultimately, this study contributes to the development of a computational framework for evaluating the spatial potential of urban regeneration interventions. By integrating structural indicators into the planning process, urban policymakers and designers can more effectively prioritize projects that are not only spatially strategic but also capable of catalyzing broader socio-economic revitalization. This

approach supports a more systematic, evidence-based strategy for urban transformation, particularly in cities facing the complex challenges of post-industrial transition.

Building on these findings, the study proposes a computational framework that integrates spatial analysis methods to assess the capacity of urban structures to support regeneration outcomes. By operationalizing measurable indicators through GIS, space syntax, and network analysis, the framework enables a systematic evaluation of spatial potential. This approach facilitates evidence-based decision-making for prioritizing regeneration projects in complex urban contexts.

FINAL REMARKS

The paper aligns closely with the objectives of the COST Action CA23145 (CitySenZ), which focuses on understanding and enhancing urban ambiances through interdisciplinary and human-centered approaches. While CitySenZ emphasizes the sensory, experiential, and socio-cultural dimensions of urban life, the paper contributes a complementary computational perspective by proposing a framework to evaluate urban structures using spatial indicators and digital tools. This spatial logic directly supports CitySenZ's ambition to foster livable, inclusive, and emotionally resonant urban environments. By leveraging tools such as GIS, space syntax, and network analysis, the paper provides a method to assess the spatial potential of regeneration sites, which can inform ambiance-sensitive planning decisions. Moreover, the emphasis on optimizing regeneration in post-industrial cities resonates with CitySenZ's focus on European urban contexts undergoing transformation. The integration of computational analysis with urban ambiance research opens pathways for evidence-based, multi-scalar urban design that bridges structural efficiency with lived experience. Thus, the authors believe the paper offers a valuable

methodological foundation that can enhance the CitySenZ agenda by grounding ambiance research in spatial data and supporting the strategic selection of regeneration sites that are not only structurally optimal but also capable of fostering vibrant, inclusive, and sensorially rich urban atmospheres.

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